

ML MODEL HANDOFF SPECIFICATION

HealthConnect Clinic Appointment No-Show Prediction Pipeline (Week 5 Deliverable)

ARTIFACT FILE	model.joblib	AUTHOR	Kutlwano Mmoloki Gabanthate
FRAMEWORK	scikit-learn (v1.6.0)	TRACK	Data Analytics / Data Science
ALGORITHM	Logistic Regression Pipeline	DATE	September 2026

1. EXECUTIVE SUMMARY & OBJECTIVE

This document details the serialized binary classification pipeline encapsulated within `model.joblib`. Designed for the HealthConnect Clinic system, the model predicts the probability of an unexcused patient appointment no-show (`appointment_outcome = No-Show`). Downstream ML engineering teams and backend services can load this pipeline directly to ingest raw features and output calibrated risk scores (0.0 to 1.0) for operational triage.

2. MODEL ARCHITECTURE & PIPELINE SCHEMA

The `.joblib` artifact contains a complete scikit-learn Pipeline bundling data preprocessing and classification into a single callable unit. Incoming payloads require no manual scaling or encoding.

FEATURE NAME	TYPE	ENCODING / SCALING	DESCRIPTION
booking_lead_days	NUMERIC	StandardScaler	Days elapsed between booking and appointment date (≥ 0).
prior_no_show_ratio	NUMERIC	StandardScaler	Ratio of historical no-shows to total prior appointments.
waiting_time_minutes	NUMERIC	StandardScaler	Expected or historic waiting time at clinic in minutes.
distance_to_clinic_km	NUMERIC	StandardScaler	Distance from patient home address to healthcare facility.
age	NUMERIC	StandardScaler	Patient age in years.
gender	CATEGORICAL	OneHotEncoder	Patient gender (e.g., 'Male', 'Female').
appointment_type	CATEGORICAL	OneHotEncoder	Clinic specialty (e.g., 'General Practice', 'Cardiology').
reminder_channel	CATEGORICAL	OneHotEncoder	Outreach channel utilized ('SMS', 'Email', 'WhatsApp', 'None').

3. VALIDATION PERFORMANCE METRICS

Evaluated on a 20% holdout test dataset (stratified split, N=1,000):

0.7680

ROC-AUC SCORE

73.5%

NO-SHOW RECALL

68.2%

PRECISION

71.4%

OVERALL ACCURACY

4. OPERATIONAL RISK THRESHOLDS

Risk Scoring Matrix for Automated Triage

- **High Risk (≥ 0.60):** Flag for immediate front-desk phone calls & double-booking protocols.
- **Medium Risk (0.35 – 0.59):** Trigger automated WhatsApp / SMS confirmation sequence.
- **Low Risk (< 0.35):** Standard scheduling queue with standard reminder.

5. PYTHON INTEGRATION CODE SNIPPET

```
import joblib
import pandas as pd

# Load serialized pipeline artifact
pipeline = joblib.load('model.joblib')

# Ingest raw JSON payload directly matching schema
raw_payload = pd.DataFrame([{'booking_lead_days': 14,
                              'prior_no_show_ratio': 0.25,
                              'waiting_time_minutes': 30,
                              'distance_to_clinic_km': 8.5,
                              'age': 38,
                              'gender': 'Female',
                              'appointment_type': 'Cardiology',
                              'reminder_channel': 'SMS'}])

# Inference execution
risk_score = pipeline.predict_proba(raw_payload)[0, 1][0]
print(f"Calculated No-Show Probability: {risk_score:.4f}")
```